

Row Echelon Form - 20 Example Problems with Step-by-Step Solutions

Problem 1 (Style: B.S. Grewal)

Find the row echelon form of the matrix:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{pmatrix}$$

Solution: Step 1: Start with the given matrix

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{pmatrix}$$

Step 2: Make zeros below the first pivot (1) $R_2 \rightarrow R_2 - 2R_1$:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 3 & 6 & 10 \end{pmatrix}$$

$R_3 \rightarrow R_3 - 3R_1$:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Step 3: Make zeros below the second pivot (1 in position (2,3)) $R_3 \rightarrow R_3 - R_2$:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Answer: Row echelon form is $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

Problem 2 (Style: R.K. Jain & S.R.K. Iyengar)

Reduce to row echelon form:

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 0 & 1 & 2 \\ 3 & 1 & 4 & 6 \end{pmatrix}$$

Solution: Step 1: Make the first element of first row = 1 $R_1 \rightarrow \frac{1}{2}R_1$:

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{3}{2} & 2 \\ 1 & 0 & 1 & 2 \\ 3 & 1 & 4 & 6 \end{pmatrix}$$

Step 2: Make zeros below first pivot $R_2 \rightarrow R_2 - R_1$:

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{3}{2} & 2 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & 0 \\ 3 & 1 & 4 & 6 \end{pmatrix}$$

$R_3 \rightarrow R_3 - 3R_1$:

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{3}{2} & 2 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & 0 \end{pmatrix}$$

Step 3: Make second pivot = 1 $R_2 \rightarrow -2R_2$:

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{3}{2} & 2 \\ 0 & 1 & 1 & 0 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & 0 \end{pmatrix}$$

Step 4: Make zeros below second pivot $R_3 \rightarrow R_3 + \frac{1}{2}R_2$:

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{3}{2} & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Answer: Row echelon form is $\begin{pmatrix} 1 & \frac{1}{2} & \frac{3}{2} & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

Problem 3 (Style: B.S. Grewal)

Find the row echelon form:

$$A = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 2 & 1 & 3 & 3 \\ 1 & 0 & 1 & 2 \\ 3 & 2 & 5 & 4 \end{pmatrix}$$

Solution: Step 1: Make zeros below first pivot $R_2 \rightarrow R_2 - 2R_1$:

$$\begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & -1 & -1 & 1 \\ 1 & 0 & 1 & 2 \\ 3 & 2 & 5 & 4 \end{pmatrix}$$

$R_3 \rightarrow R_3 - R_1$:

$$\begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & -1 & -1 & 1 \\ 0 & -1 & -1 & 1 \\ 3 & 2 & 5 & 4 \end{pmatrix}$$

$R_4 \rightarrow R_4 - 3R_1$:

$$\begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & -1 & -1 & 1 \\ 0 & -1 & -1 & 1 \\ 0 & -1 & -1 & 1 \end{pmatrix}$$

Step 2: Make second pivot positive $R_2 \rightarrow -R_2$:

$$\begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & -1 & -1 & 1 \\ 0 & -1 & -1 & 1 \end{pmatrix}$$

Step 3: Make zeros below second pivot $R_3 \rightarrow R_3 + R_2$ and $R_4 \rightarrow R_4 + R_2$:

$$\begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Answer: Row echelon form is $\begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

Problem 4 (Style: R.K. Jain & S.R.K. Iyengar)

Transform to row echelon form:

$$A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

Solution: Step 1: Since first element is 0, interchange rows $R_1 \leftrightarrow R_2$:

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 2 & 1 & 3 \end{pmatrix}$$

Step 2: Make zeros below first pivot $R_3 \rightarrow R_3 - 2R_1$:

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & -3 & 1 \end{pmatrix}$$

Step 3: Make zeros below second pivot $R_3 \rightarrow R_3 + 3R_2$:

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 7 \end{pmatrix}$$

Answer: Row echelon form is $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 7 \end{pmatrix}$

Problem 5 (Style: B.S. Grewal)

Reduce the following matrix to row echelon form:

$$A = \begin{pmatrix} 1 & -1 & 2 & -1 \\ 2 & -2 & 3 & 2 \\ 1 & -1 & 1 & 3 \end{pmatrix}$$

Solution: Step 1: Make zeros below first pivot $R_2 \rightarrow R_2 - 2R_1$:

$$\begin{pmatrix} 1 & -1 & 2 & -1 \\ 0 & 0 & -1 & 4 \\ 1 & -1 & 1 & 3 \end{pmatrix}$$

$R_3 \rightarrow R_3 - R_1$:

$$\begin{pmatrix} 1 & -1 & 2 & -1 \\ 0 & 0 & -1 & 4 \\ 0 & 0 & -1 & 4 \end{pmatrix}$$

Step 2: Make second pivot positive $R_2 \rightarrow -R_2$:

$$\begin{pmatrix} 1 & -1 & 2 & -1 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & -1 & 4 \end{pmatrix}$$

Step 3: Make zeros below second pivot $R_3 \rightarrow R_3 + R_2$:

$$\begin{pmatrix} 1 & -1 & 2 & -1 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Answer: Row echelon form is $\begin{pmatrix} 1 & -1 & 2 & -1 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

Problem 6 (Style: R.K. Jain & S.R.K. Iyengar)

Find row echelon form of:

$$A = \begin{pmatrix} 2 & 4 & 6 & 8 \\ 1 & 3 & 5 & 7 \\ 3 & 5 & 7 & 9 \end{pmatrix}$$

Solution: Step 1: Make first element = 1 $R_1 \rightarrow \frac{1}{2}R_1$:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 5 & 7 \\ 3 & 5 & 7 & 9 \end{pmatrix}$$

Step 2: Make zeros below first pivot $R_2 \rightarrow R_2 - R_1$:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 3 & 5 & 7 & 9 \end{pmatrix}$$

$R_3 \rightarrow R_3 - 3R_1$:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & -1 & -2 & -3 \end{pmatrix}$$

Step 3: Make zeros below second pivot $R_3 \rightarrow R_3 + R_2$:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Answer: Row echelon form is $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

Problem 7 (Style: B.S. Grewal)

Transform to row echelon form:

$$A = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 5 & 1 \\ 1 & 1 & 6 & -2 \end{pmatrix}$$

Solution: Step 1: Make zeros below first pivot $R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 3R_1, R_4 \rightarrow R_4 - R_1$:

$$\begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & -1 & 3 & -2 \\ 0 & -2 & 8 & -8 \\ 0 & -1 & 7 & -5 \end{pmatrix}$$

Step 2: Make second pivot positive $R_2 \rightarrow -R_2$:

$$\begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & -3 & 2 \\ 0 & -2 & 8 & -8 \\ 0 & -1 & 7 & -5 \end{pmatrix}$$

Step 3: Make zeros below second pivot $R_3 \rightarrow R_3 + 2R_2, R_4 \rightarrow R_4 + R_2$:

$$\begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 2 & -4 \\ 0 & 0 & 4 & -3 \end{pmatrix}$$

Step 4: Make zeros below third pivot $R_4 \rightarrow R_4 - 2R_3$:

$$\begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 2 & -4 \\ 0 & 0 & 0 & 5 \end{pmatrix}$$

Answer: Row echelon form is $\begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 2 & -4 \\ 0 & 0 & 0 & 5 \end{pmatrix}$

Problem 8 (Style: R.K. Jain & S.R.K. Iyengar)

Find the row echelon form:

$$A = \begin{pmatrix} 0 & 0 & 2 & 1 \\ 1 & 1 & 1 & 2 \\ 2 & 2 & 0 & 3 \end{pmatrix}$$

Solution: Step 1: Interchange rows to get non-zero element in first position $R_1 \leftrightarrow R_2$:

$$\begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 2 & 2 & 0 & 3 \end{pmatrix}$$

Step 2: Make zeros below first pivot $R_3 \rightarrow R_3 - 2R_1$:

$$\begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -2 & -1 \end{pmatrix}$$

Step 3: Make zeros below second pivot $R_3 \rightarrow R_3 + R_2$:

$$\begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Answer: Row echelon form is $\begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

Problem 9 (Style: B.S. Grewal)

Reduce to row echelon form:

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 6 & 2 & 5 & 7 \\ 9 & 3 & 8 & 10 \end{pmatrix}$$

Solution: Step 1: Make first element = 1 $R_1 \rightarrow \frac{1}{3}R_1$:

$$\begin{pmatrix} 1 & \frac{1}{3} & \frac{2}{3} & \frac{4}{3} \\ 6 & 2 & 5 & 7 \\ 9 & 3 & 8 & 10 \end{pmatrix}$$

Step 2: Make zeros below first pivot $R_2 \rightarrow R_2 - 6R_1$:

$$\begin{pmatrix} 1 & \frac{1}{3} & \frac{2}{3} & \frac{4}{3} \\ 0 & 0 & 1 & -1 \\ 9 & 3 & 8 & 10 \end{pmatrix}$$

$$R_3 \rightarrow R_3 - 9R_1:$$

$$\begin{pmatrix} 1 & \frac{1}{3} & \frac{2}{3} & \frac{4}{3} \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 2 & -2 \end{pmatrix}$$

$$\text{Step 3: Make zeros below second pivot } R_3 \rightarrow R_3 - 2R_2:$$

$$\begin{pmatrix} 1 & \frac{1}{3} & \frac{2}{3} & \frac{4}{3} \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Answer: Row echelon form is $\begin{pmatrix} 1 & \frac{1}{3} & \frac{2}{3} & \frac{4}{3} \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

Problem 10 (Style: R.K. Jain & S.R.K. Iyengar)

Transform the matrix to row echelon form:

$$A = \begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 2 & 7 & 2 & 5 \\ 1 & 4 & 2 & 3 \end{pmatrix}$$

Solution: Step 1: Make zeros below first pivot $R_3 \rightarrow R_3 - 2R_1, R_4 \rightarrow R_4 - R_1:$

$$\begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 \end{pmatrix}$$

Step 2: Make zeros below second pivot $R_3 \rightarrow R_3 - R_2, R_4 \rightarrow R_4 - R_2:$

$$\begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Answer: Row echelon form is
$$\begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Problem 11 (Style: B.S. Grewal)

Find row echelon form:

$$A = \begin{pmatrix} 2 & -1 & 3 & 2 \\ 4 & -2 & 5 & 3 \\ 6 & -3 & 7 & 4 \end{pmatrix}$$

Solution: Step 1: Make first element = 1 $R_1 \rightarrow \frac{1}{2}R_1$:

$$\begin{pmatrix} 1 & -\frac{1}{2} & \frac{3}{2} & 1 \\ 4 & -2 & 5 & 3 \\ 6 & -3 & 7 & 4 \end{pmatrix}$$

Step 2: Make zeros below first pivot $R_2 \rightarrow R_2 - 4R_1$:

$$\begin{pmatrix} 1 & -\frac{1}{2} & \frac{3}{2} & 1 \\ 0 & 0 & -1 & -1 \\ 6 & -3 & 7 & 4 \end{pmatrix}$$

$R_3 \rightarrow R_3 - 6R_1$:

$$\begin{pmatrix} 1 & -\frac{1}{2} & \frac{3}{2} & 1 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -2 & -2 \end{pmatrix}$$

Step 3: Make second pivot positive $R_2 \rightarrow -R_2$:

$$\begin{pmatrix} 1 & -\frac{1}{2} & \frac{3}{2} & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -2 & -2 \end{pmatrix}$$

Step 4: Make zeros below second pivot $R_3 \rightarrow R_3 + 2R_2$:

$$\begin{pmatrix} 1 & -\frac{1}{2} & \frac{3}{2} & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Answer: Row echelon form is $\begin{pmatrix} 1 & -\frac{1}{2} & \frac{3}{2} & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

Problem 12 (Style: R.K. Jain & S.R.K. Iyengar)

Reduce the following matrix to row echelon form:

$$A = \begin{pmatrix} 1 & 0 & 2 & -1 \\ 3 & 1 & 7 & -2 \\ 2 & 1 & 5 & -1 \\ 4 & 1 & 9 & -3 \end{pmatrix}$$

Solution: Step 1: Make zeros below first pivot $R_2 \rightarrow R_2 - 3R_1, R_3 \rightarrow R_3 - 2R_1, R_4 \rightarrow R_4 - 4R_1$:

$$\begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

Step 2: Make zeros below second pivot $R_3 \rightarrow R_3 - R_2, R_4 \rightarrow R_4 - R_2$:

$$\begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Answer: Row echelon form is $\begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

Problem 13 (Style: B.S. Grewal)

Find the row echelon form:

$$A = \begin{pmatrix} 0 & 2 & 1 & 3 \\ 0 & 4 & 2 & 7 \\ 1 & 3 & 2 & 5 \end{pmatrix}$$

Solution: Step 1: Interchange rows to get non-zero element in first position $R_1 \leftrightarrow R_3$:

$$\begin{pmatrix} 1 & 3 & 2 & 5 \\ 0 & 4 & 2 & 7 \\ 0 & 2 & 1 & 3 \end{pmatrix}$$

Step 2: Make second pivot = 1 $R_2 \rightarrow \frac{1}{4}R_2$:

$$\begin{pmatrix} 1 & 3 & 2 & 5 \\ 0 & 1 & \frac{1}{2} & \frac{7}{4} \\ 0 & 2 & 1 & 3 \end{pmatrix}$$

Step 3: Make zeros below second pivot $R_3 \rightarrow R_3 - 2R_2$:

$$\begin{pmatrix} 1 & 3 & 2 & 5 \\ 0 & 1 & \frac{1}{2} & \frac{7}{4} \\ 0 & 0 & 0 & -\frac{1}{2} \end{pmatrix}$$

Answer: Row echelon form is $\begin{pmatrix} 1 & 3 & 2 & 5 \\ 0 & 1 & \frac{1}{2} & \frac{7}{4} \\ 0 & 0 & 0 & -\frac{1}{2} \end{pmatrix}$

Problem 14 (Style: R.K. Jain & S.R.K. Iyengar)

Transform to row echelon form:

$$A = \begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 4 & -1 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Solution: The given matrix is already in row echelon form since:

- All non-zero rows are above any rows of all zeros
- Each leading entry of a row is in a column to the right of the leading entry of the row above it
- All entries in a column below a leading entry are zeros

Answer: The matrix is already in row echelon form:
$$\begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 4 & -1 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Problem 15 (Style: B.S. Grewal)

Reduce to row echelon form: $A = \begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 4 & -1 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$